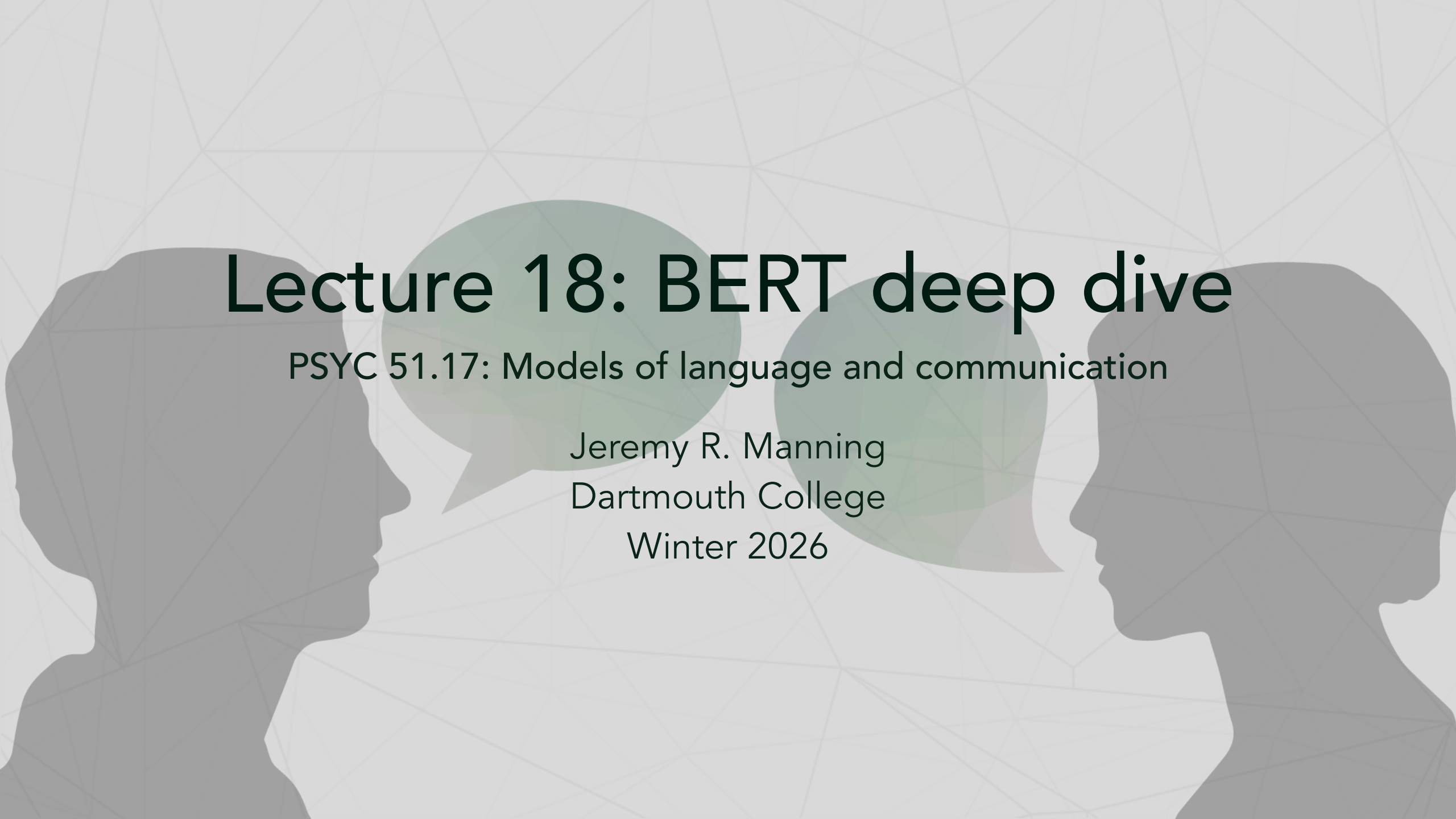




Lecture 18: BERT deep dive

PSYC 51.17: Models of language and communication

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Learning objectives

By the end of this lecture, you will be able to...

1. Explain the ONE architectural change that makes BERT bidirectional
2. Analyze concrete examples where unidirectional models fail
3. Describe what different BERT layers and attention heads learn
4. Run BERT code to explore masked predictions and attention patterns
5. Connect BERT's design to cognitive and linguistic insights

What you already know

Recap from lectures 12 and 15

In lecture 12, we introduced BERT basics: Masked Language Modeling (MLM) with the 80/10/10 strategy, Next Sentence Prediction (NSP), and the input representation (token + segment + position embeddings). In lecture 15, we built the full transformer architecture, including Q/K/V attention and causal masking. Today we go deeper into WHY these design choices matter and WHAT BERT actually learns.

Breaking the causal mask

The key difference is the mask

Recall that attention scores are calculated as:

- GPT: $A_{\text{GPT}} = \text{softmax} \left(\frac{QK^T}{\sqrt{H}} + M_{\text{causal}} \right)$
- BERT: $A_{\text{BERT}} = \text{softmax} \left(\frac{QK^T}{\sqrt{H}} \right)$

Bidirectionality: One line of math — removing the causal mask — transforms a next-word predictor into a bidirectional understanding machine.

Trade-off

BERT cannot generate text autoregressively because it "sees the future" during processing. It is an encoder, not a decoder.

Why bidirectionality matters

Contextual ambiguity

Consider these Winograd-style examples where left-context alone is insufficient:

- "The trophy didn't fit in the suitcase because **it** was too [big/small]" — "it" refers to trophy or suitcase depending on the final word.
- "The animal didn't cross the street because **it** was too [wide/tired]" — need right context to resolve "it".

The BERT advantage

A left-to-right model processes "it" before seeing "big" vs "small" — it must guess. BERT reads both directions and never has to guess.

The Winograd schema challenge

Winograd schemas

Pronoun resolution challenges that test real-world understanding. Originally proposed by Terry Winograd (1972), formalized as a benchmark by Levesque et al. (2012).

Ambiguous pronouns:

- "The city councilmen refused the demonstrators a permit because **they** feared violence." (they = councilmen)
- "The city councilmen refused the demonstrators a permit because **they** advocated violence." (they = demonstrators)

Performance

BERT-large was among the first models to approach human performance on Winograd schemas.

What fill-in-the-blank teaches

Intuition for MLM

Filling in blanks forces BERT to learn grammar ("The [MASK] barked" → dog), semantics ("She [MASK] the exam" → passed/failed), and world knowledge ("The capital of France is [MASK]" → Paris).

Self-supervised curriculum

Think of MLM as a self-supervised curriculum: every masked position is a test question, and the surrounding text is the study material.

Try it: what does BERT predict?

Code example

```
1 from transformers import pipeline
2 fill_mask = pipeline("fill-mask", model="bert-base-uncased")
3 fill_mask("The [MASK] barked at the mailman.")
4 # → [{'token_str': 'dog', 'score': 0.65}, ... ]
5
6 fill_mask("The capital of Japan is [MASK].")
7 # → [{'token_str': 'tokyo', 'score': 0.85}, ... ]
```


BERT reveals its biases

Exposing training data biases

```
1 fill_mask("The nurse said [MASK] would be right back.")
2 # → 'she' (0.58), 'he' (0.25), 'i' (0.06)
3 fill_mask("The doctor said [MASK] would be right back.")
4 # → 'he' (0.55), 'she' (0.22), 'i' (0.08)
```

Feature and bug

BERT learned gender stereotypes from its training data. This is useful for studying bias, but dangerous if deployed uncritically.

What attention heads specialize in

Linguistic specialization

Clark et al. (2019) discovered that specific BERT attention heads specialize in linguistic relations:

- **Head 8-10:** Direct objects (verb → object)
- **Head 7-6:** Possessive pronouns (noun → its/his/her)
- **Head 5-4:** Coreference (pronoun → antecedent)

Emergent properties

No one programmed these specializations — they emerged from masked language modeling alone.

BERT learns a linguistic pipeline

Layer-wise processing

Tenney et al. (2019) showed BERT's layers form a processing pipeline:

1. **Lower layers (1–4):** Surface features — POS tags, word boundaries
2. **Middle layers (5–8):** Syntax — parse trees, dependency relations
3. **Upper layers (9–12):** Semantics — word sense disambiguation, coreference

Classical NLP

This mirrors the classical NLP pipeline that used to require separate hand-crafted systems for each level.

Embeddings change across layers

Word sense transformation

Consider the word "cells" in: "The cells in the prison..." vs "The cells in the body..."

- **Layer 1:** Both "cells" have similar representations (surface form).
- **Layer 6:** Syntactic context starts differentiating them.
- **Layer 12:** Completely different vectors (prison vs biology).

Measurement

Cosine similarity between the two "cells" vectors drops from ~ 0.95 (layer 1) to ~ 0.45 (layer 12).

Visualizing attention patterns

Bertviz exploration

In the companion notebook, you'll use `bertviz` to see attention patterns. These visualizations show how tokens attend to each other across different heads and layers. Try sentences with pronouns — which attention head links "it" to its antecedent?

Attention maps

Attention maps often show "ribbon" diagrams or heatmaps where lines connect tokens, indicating the strength of the attention weights.

BERT-base vs BERT-large

Architecture comparison

Component	BERT-base	BERT-large
Layers	12	24
Hidden size	768	1024
Attention heads	12	16
Parameters	110M	340M
Max sequence	512	512

Usage

BERT-base fits on a single GPU and runs in seconds. BERT-large is ~3x slower but excels on complex reasoning tasks like Winograd schemas.

The pre-training data question

What's in the data?

BERT was trained on 3.3B words of BooksCorpus + Wikipedia:

- Primarily English, formal writing.
- Over-represents certain demographics and viewpoints.
- No social media, no conversations, no code.
- Cutoff: 2018 (no knowledge of recent events).

Data shapes the model

BERT "knows" what its training data contained. Its biases, gaps, and capabilities all trace back to these 3.3 billion words.

From understanding to doing

Pre-train then fine-tune

- **Pre-training:** Expensive, one-time, learns general language understanding.
- **Fine-tuning:** Cheap, per-task, adds a task-specific head.

Transfer learning

The key insight: language understanding transfers. A model that can fill in blanks has learned enough about language to excel at classification, QA, NER, and more.

Fine-tuning in 5 lines

Sentiment analysis pipeline

```
1 from transformers import pipeline
2 classifier = pipeline("sentiment-analysis")
3 classifier("This movie was absolutely wonderful!")
4 # → [{'label': 'POSITIVE', 'score': 0.9998}]
5 classifier("The plot was confusing and dull.")
6 # → [{'label': 'NEGATIVE', 'score': 0.9987}]
```

Abstraction

Under the hood, this is a BERT model fine-tuned on sentiment data. The pipeline abstracts away tokenization, model loading, and inference.

BERT's lasting impact

Historical timeline

- **2013:** Word2Vec — static word embeddings
- **2018 Feb:** ELMo — contextual, but RNN-based
- **2018 Oct:** BERT — bidirectional transformers
- **2019+:** RoBERTa, ALBERT, DistilBERT, ELECTRA, DeBERTa

The paradigm shift

BERT established the "pre-train then fine-tune" paradigm that now dominates NLP. Even GPT-4 and Claude use this approach — they just use different architectures and much more data.

BERT and the brain

Neural parallels

Neuroscience studies have found striking parallels between BERT's internal representations and human brain activity during language processing.

Cognitive insights:

- BERT's layer progression mirrors the temporal cascade of language processing in the brain.
- Middle BERT layers best predict fMRI activity in language regions.
- The N400 ERP component correlates with BERT's prediction confidence.

Computational solutions

Does BERT "understand" language like we do? Probably not — but it may have discovered similar computational solutions to the same problem.

Questions to think about

Discussion questions

1. Why can't BERT generate text like GPT? Is understanding fundamentally different from generation?
2. If BERT learns linguistic structure from fill-in-the-blank alone, what does that tell us about how much structure is in language itself?
3. Could you train a BERT-like model on music, images, or DNA sequences? What would "masking" mean in those domains?
4. BERT reveals gender biases. Should we fix the model, fix the data, or both?

References

Further reading

- Devlin et al. (2019, NAACL) — BERT original paper
- Tenney et al. (2019, ACL) — "BERT rediscovers the classical NLP pipeline"
- Clark et al. (2019, BlackboxNLP) — "What does BERT look at?"
- Levesque et al. (2012) — Winograd Schema Challenge
- Schrimpf et al. (2021, PNAS) — Neural language models and the brain
- HuggingFace Course, Chapter 1 — Practical introduction
- Jay Alamar: The Illustrated BERT

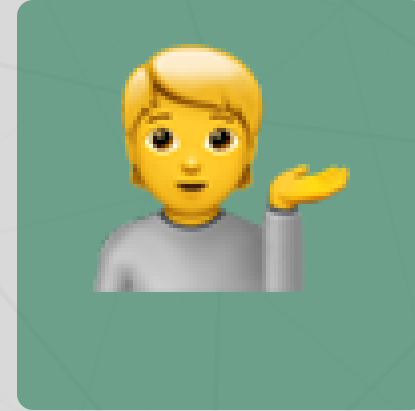
Questions?



Email



Discord



Office hours

Up next...

BERT variants: how RoBERTa, ALBERT, DistilBERT, and ELECTRA improved on the original. 📖 Companion notebook